

Unlabeled batch X_t



Classifier

f_{θ_t}

θ_t : online update

Current outputs

logits $\mathbf{o}_{t,i} \rightarrow$ prediction $\mathbf{p}_{t,i}$

features $\mathbf{z}_{t,i}$ & $\mathbf{v}_{t,i}$

Regional evidence

RE $\ell_{RE,i}$ + PLPD
RI $\ell_{RI,i}$ selected samples

regional objective

Sparse external queries $\rightarrow$ dense, persistent guidance

1 Sparse semantic anchors

Candidate window $\mathcal{W}_t$



vector values

$x_{t,i}$ $v_{t,i}$ $p_{t,i}$

scalar values

$\ell_{RE,t,i}$ $\ell_{RI,t,i}$

$\phi_{t,i}$ $m_{t,i}$ $q_{t,i}$

Anchor ranking



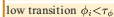
reliable $q_i=1$

large margin $m_i > \tau_m$

low RE/RI $\ell_{RE,i} < \tau_{RE}$

$\ell_{RI,i} < \tau_{RI}^{anc}$

low transition $\phi_i < \tau_\phi$



rank-1

Frozen MLLM



Frozen text encoder

Refresh if any

No anchors

Scheduled refresh

Few memory matches

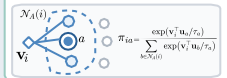
Visual drift

2 Semantic propagation

Sparse anchor bank $\mathcal{A}_t$

$\mathbf{u}_a$ $\mathbf{e}_a$ $\mathbf{c}_a$ κ_a

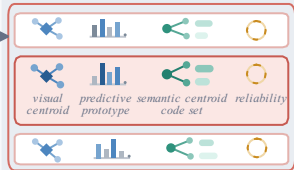
Visual KNN $|\mathcal{N}_A(i)| = \min(K_a, |\mathcal{A}_t|)$



Type-specific propagation



3 Visual-semantic prototype memory



Visual-semantic prototype retrieval

READ
matched prototype
prototype consistency loss

WRITE
Reliable observations
cluster update

4 Joint online adaptation

$\mathcal{L}_{\text{region}}$

$\mathcal{L}_{\text{proto}}$

One gradient step
normalization-affine θ_t only